

# Feature Alignment: Attack Characteristics → Features

**Group:** MITM / Evil Twin Attacks

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**Attack Category:** Man-in-the-Middle – Evil Twin / Rogue Access Point

**MITRE ATT&CK:** T1557 (Adversary-in-the-Middle), T1040 (Network Sniffing), T1185 (Browser Session Hijacking)

This document maps the technical characteristics of the Evil Twin / rogue-AP attack (defined in Period 1) to the concrete features and raw signals identified across the group's four Period-2 papers. It forms the basis for the harmonized feature schema we will build in Period 4.

## 1. Papers Reviewed (Period 2)

- **P1 – Shaked:** da Silva et al. 2023, LightGBM on AWID3. Feature selection: 16 expert features (Chatzoglou et al.), then removed constant & Pearson > 0.95 correlated columns.
- **P2 – Itay:** Kamble & Kshirsagar 2023, Random Forest + Gini Index on AWID3. Gini reduced 207 → 25 features (beat IG / ANOVA, lowest build time).
- **P3 – Michael:** Chauhan 2023, CART / FCNN / Ensemble on AWID3. ANOVA F-test reduced 34 → 13 columns (12 features + label).
- **P4 – Iftach:** Paula et al. 2024, One-Class SVM on the RRAD rogue-AP dataset. Domain-driven 20 features (9 software + 11 hardware).

## 2. Evil Twin Attack Characteristics (from Period 1)

- Rogue Wi-Fi access points impersonating legitimate SSIDs, with duplicate SSIDs on different MAC / BSSID addresses.
- Sudden signal-strength anomalies and suspicious access-point switching.
- Intercepted authentication traffic, DNS spoofing / redirection, and unusual packet-forwarding behaviour.
- Session hijacking, credential theft, fake captive portals, and abnormal beacon-frame timing.

## 3. Attack Characteristic → Feature Mapping

| Attack Characteristic                    | Mapped Feature(s) / Observable Signal                                     | Source / Layer       | Papers     |
|------------------------------------------|---------------------------------------------------------------------------|----------------------|------------|
| Rogue AP impersonating SSID (RF anomaly) | radiotap.dbm_antsignal; Short GI 20MHz, supported data rates, Receive NDP | Radiotap / PHY       | P1, P4     |
| Duplicate SSID, different MAC / BSSID    | wlan.fc.type, wlan.fc.subtype, wlan.fc.ds; hardware fingerprint features  | MAC header           | P1, P2, P4 |
| Intercepted / relayed traffic            | frame.len, wlan.duration, Cap MPDU                                        | MAC / frame metadata | P1, P2, P4 |
| Weak / mismatched encryption             | wlan.fc.protected (most discriminative categorical feature)               | MAC frame-control    | P1         |
| Abnormal beacon / timing                 | radiotap.present.tsft, radiotap.channel.freq, channel.flags.cck/ofdm      | Radiotap / timing    | P1         |
| Management-frame manipulation            | wlan.fc.retry, wlan.fc.frag, wlan.fc.pwrmtg, wlan.fc.moredata             | MAC frame-control    | P1, P2, P3 |
| Session hijacking / credential theft     | GI-selected app-layer features (25); ANOVA-selected app features          | Application layer    | P2, P3     |
| Selective targeting / host anomaly       | One-class 'normal AP' profile; hard-to-spoof hardware features            | PHY + behavioural    | P4         |

## 4. Notes Toward a Harmonized Schema (Period 4)

- **Strong convergence:** three of four papers (P1–P3) use AWID3, so MAC-layer frame-control (wlan.fc.\*) and radiotap signal/timing features are directly shareable across datasets.
- **Cross-dataset gap:** P4 uses RRAD with hardware/PHY features (Short GI, Cap MPDU, NDP) absent from AWID3 – these need derivation, omission, or a documented justification in the unified schema.

- **Candidate minimal feature set:** across the AWID3 papers, wlan.fc.protected plus the numerical trio radiotap.dbm\_antsignal, frame.len, wlan.duration repeatedly emerge as the core evil-twin discriminators.